Software Requirements Specification

Worked Example — StudyMate (Instructor's Teaching Demo)

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

Field	This example
Team name	Strategic
Project title	Elec-Specify — Dormitory Electricity Cost Management System
SDLC chosen in M1 + reason	**Agile / Scrum** — Flexible step-by-step development that allows restructuring based on user feedback, provides control over logical/computational errors, and maintains project transparency.
Date / version	v1.0 · Week 3 SRS
GitHub repo	https://github.com/Jirapat693150/-2

1. Introduction

1.1 Purpose and scope

Students juggle classes, deadlines, group work, and club activities across scattered notes, chat messages, and memory — and deadlines get missed not because students don't care, but because nothing keeps the next deadline in front of them at the right moment. StudyMate is a small assistant that stores a student's tasks with due dates, reminds them one day before a deadline, and shows the coming week at a glance, so that a task never disappears until it is either done or has been actively planned around. The MVP is deliberately small: four core stories (add/edit a task, mark it done, get a reminder, see the week) plus one AI-assisted convenience feature (turn a typed natural-language message into a structured task). Success looks like this: a student stops missing deadlines because they forgot, not because the system failed to warn them in time.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Dormitory Renter / Student (Patrpor Wutisutr, Waratit Pongwiwat)	primary user	Clear visibility of daily/monthly electricity costs and individual appliance consumption breakdown.	Inaccurate calculations leading to unexpected bill spikes; overly complex manual input forms.
Home / Property Leader (Saroj Kruengsai)	secondary stakeholder	A system to manage house/room electricity usage habits and budget effectively	Unreliable estimates that do not match actual utility bills.
Dormitory Landlord / Utility Provider	institution or supplier	Fair and transparent billing understanding by renters without dispute.	Misunderstandings regarding electricity rate tariffs vs. app calculations.

2. Overall description

2.1 Product context — what it does / does not

StudyMate keeps one list of tasks per student and turns “1 day before due” into a reminder. Everything below is the whole MVP for this term — nothing else is implied.

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
• **FR-1:** User Profile & Presets • **FR-2:** Computing System (Cost Calculation Engine) • **FR-3:** User Guide & Help Documentation • **FR-4:** Result Declaration & Cost Breakdown • **FR-5:** Usage History & Session Persistence	• Automatic IoT hardware integration / smart plug syncing. • Direct electricity bill payment processing. • Real-time smart meter hardware tracking. • Automated messaging/notifications to landlords.

2.2 Assumptions and constraints

- Assumption: every student has a smartphone and a university account used to sign in.
- Constraint: one LLM API call per natural-language add — no model training, no fine-tuning.
- Constraint: Thai and English UI text; no paid API beyond the single LLM call above.
- Constraint: personal data — we store task title, due date, and course name only; we do not store message content from any chat app.
- Constraint: 10-week build window shared with four other milestones (M1–M5).

3. Functional requirements

Format: As a <role>, I want <goal>, so that <benefit>. Every story below traces to a pain observed in a (simulated) student interview, and every acceptance criterion is something a test could actually check.

FR-1 — User Profile & Appliance Presets

- **User story: As a dormitory renter, I want to create a profile and select appliance presets (e.g., fan, refrigerator, light bulb), so that I don't have to re-enter wattage and baseline info every time I calculate my bill.**
- **Pain this traces to: Renters forget which appliances they run and get fatigued by filling out redundant forms repeatedly.**
- **Acceptance Criteria:**
    1. *Given* a user creating or editing their profile, *when* they select an appliance preset (e.g., Standard Fan - 50W), *then* the system populates the wattage field automatically.
    2. *Given* a saved profile, *when* the user re-opens the app, *then* their saved list of appliances remains loaded.

FR-2 — Computing System (Energy & Cost Calculation)

- **User story: As a dormitory renter, I want the system to calculate daily and monthly electricity costs based on my usage hours and room electricity rates, so that I know my total expected expense beforehand.**
- **Pain this traces to: Renters run high-wattage appliances simultaneously without knowing the financial impact until the bill arrives.**
- **Acceptance Criteria:**
    1. *Given* valid inputs (wattage, hours per day, unit price), *when* the calculation is executed, *then* the system calculates total kWh and total cost accurately within less than 1 second.
    2. *Given* an invalid input (e.g., entering 25 hours of daily usage for a single device), *when* the user attempts to compute, *then* the system blocks computation and displays a validation error message.

FR-3 — User Guide & Help Documentation

- **User story: As a user, I want an easily accessible user guide within the app, so that I can learn how to input appliance details correctly and interpret calculation results accurately.**
- **Pain this traces to: Users entering incorrect usage times due to confusion, causing human error in calculations.**
- **Acceptance Criteria:**
    1. *Given* a user navigating to the User Guide page, *when* they click on any help topic (e.g., "How to find appliance wattage"), *then* step-by-step instructions with examples are displayed.
    2. *Given* a first-time user, *when* they access the main calculation form, *then* a tooltip/help button is visible next to complex input fields.

FR-4 — Result Declaration & Breakdown

- **User story: As a dormitory renter, I want to see a clear result display showing the total monthly cost alongside an appliance breakdown, so that I know exactly which appliances consume the most electricity.**
- **Pain this traces to: Renters receive a single total bill amount without knowing where most of the electricity went, making it impossible to adjust habits.**
- **Acceptance Criteria:**
    1. *Given* completed calculations from FR-2, *when* the result screen loads, *then* the total cost is declared prominently alongside a breakdown list/chart sorted by energy consumption per appliance.
    2. *Given* an updated input in FR-2, *when* recalculated, *then* the result display updates immediately without showing stale data.

FR-5 — History & Data Persistence

- **User story: As a dormitory renter, I want my calculation history saved between sessions, so that I can compare previous months' estimated usage against actual bills.**
- **Pain this traces to: Users losing past usage data and being unable to track improvements in energy consumption over time.**
- **Acceptance Criteria:**
    1. *Given* a user finishes a calculation, *when* they close and re-open the application, *then* their past calculation logs remain accessible in the history page.
    2. *Given* a user clearing their history, *when* confirmed, *then* past entries are removed and an empty-state message is shown.

4. Non-functional requirements

Forbidden words unless quantified: fast, easy, user-friendly, smart, secure. NFR-5 (security/privacy) is required for M2 — every SRS must have at least one.

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	The calculation system (FR-2 & FR-4) shall display updated total results within **1.5 seconds** after pressing "Calculate".	Benchmark test using Chrome DevTools across 5 test trials.
NFR-2	performance	A new user shall complete a profile setup and produce their first electricity estimation within **45 seconds** without external aid.	Timed usability trial with 3 test subjects.
NFR-3	usability	Saved user profile and usage history data shall persist without corruption or data loss across browser sessions / app restarts.	Perform 10 force-close and restart test scenarios; verify data integrity.

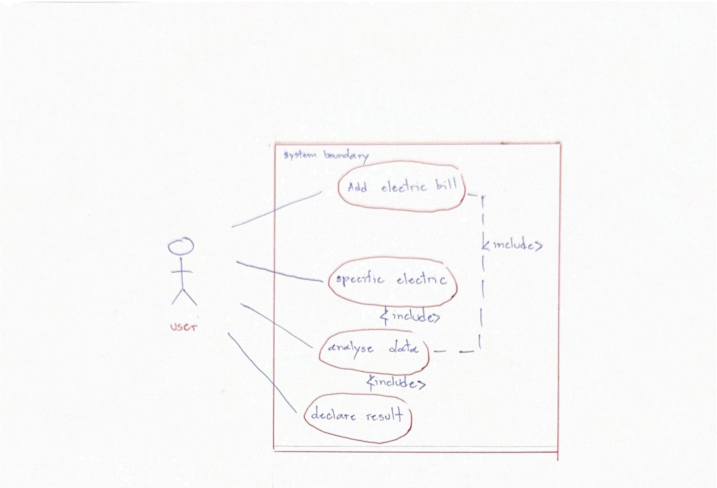
ID	Kind	Statement (testable)	How we will check
NFR-4	reliability	The computing system shall achieve 100% mathematical accuracy based on the formula: Cost = (Wattage * Hour) /1000 * Unit Rate	Unit test suite covering zero, integer, and decimal input values.
NFR-5	security / privacy (required)	User profile data and calculation logs shall be isolated; a user shall not be able to view or modify another user's profile or history.	API/Session inspection; attempt accessing restricted endpoints using another user ID.

5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1 Setup Profile & Presets	Dormitory Renter	FR-1	User configures room profile and selects preset appliances.
UC-2 Compute Electricity Cost	Dormitory Renter	FR-2	User inputs usage hours; system validates input and computes total costs.
UC-3 Access User Guide	Dormitory Renter	FR-3	User opens documentation to learn how to check device wattage.
UC-4 View Cost Result & Breakdown	Dormitory Renter	FR-4	System displays declared total result and individual device expense breakdown.
UC-5 View History Logs	Dormitory Renter	FR-5	User reviews saved calculations from previous sessions.

5.2 Use-case diagram



6. Out of scope

1. Direct integration with IoT smart plugs or external hardware meters.
2. In-app payment processing for dormitory utility bills.
3. Automated landlord-renter notification system.
4. Real-time dynamic grid tariff tracking outside of user-defined unit rates.

7. Traceability matrix (Golden Thread)

Only FR-3 is filled all the way across — that is the single thread this course follows week by week (Ch3 → Ch4/5 → Ch6 → Ch7). The other rows stay open until their week arrives, exactly like your own M2 should.

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Unaware of pre-calculated costs leading to unexpected bill spikes	FR-1 Profile & Presets	(Week 5)	(Week 6–7)	(Week 7)
Human error in daily usage input causing incorrect calculation	FR-2 Computing System	(Week 5)	(Week 6–7)	(Week 7)
Lack of instructions leading to misoperation	FR-3 User Guide	Calculation Engine + Input Validator (Ch 4–5)	"Electricity Calculation Engine" (Ch 6)	unit tests (Ch 7)
No clear breakdown of where electricity goes	FR-4 Result Declare	(Week 5)	(Week 6–7)	(Week 7)
Loss of data between sessions	FR-5 Data Persistence	(Week 5)	(Week 6–7)	(Week 7)

8. AI usage log

If this table is empty but the prose looks generated, the milestone is capped at 50% — log every prompt that drafted a story or paragraph, honestly.

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned
13 Sep 2026	Gemini	Map M1 Charter into StudyMate M2 SRS format with testable FRs/NFRs.	Complete SRS draft structured according to StudyMate sample.	Adjusted FR descriptions to strictly align with Team Strategic's M1 Charter; added human-error validation criteria to FR-2.	AI generated correct structure, but functional criteria needed specific validation rules matching our domain context.

9. Self-check

- ☑ Every FR is a user story with a benefit — all 5 FRs in §3 end in a “so that” clause tied to a real pain.
- ☑ Every FR has acceptance criteria you could actually test — 2 AC per FR, each a number, a state, or a yes/no.

- ☑ No FR is vague — no bare “fast”, “easy”, or “user-friendly” anywhere in §3.
- ☑ NFRs exist, including security / privacy — NFR-5 is a required access-control check.
- ☑ Every FR traces back to a stated problem — see “Pain this traces to” under each FR, and §7.
- ☑ Scope and out-of-scope are stated — §2.1 and §6.
- ☑ Use-case list matches the FR list — 5 use cases, one per FR (diagram added Week 4).
- ☑ AI log is honest — §8 shows exactly what was asked, drafted, and hand-corrected.

Course rule: you may use AI to draft. You must log it, verify it against a real stakeholder, and be able to explain every FR in the oral exam. Work you cannot explain earns no credit.